INDUSTRY PROJECT REPORT

HR Data Analytics

Industry Project Title	IBM HR Analytics Employee Attrition & Performance
Name of the Company	TCS
Name of the Institute	Guru Nanak College of Arts, Science & Commerce

Start Date	End Date	Total Effort (hrs.)	Project Environment	Tools used
16 April 2026	21 April 2026	8 hrs	Windows	Power BI, VS code, Jupiter Notebook,

1. Acknowledge:

I would like to express my sincere gratitude to my institute and mentors for providing me with the opportunity to work on this industry project. Their guidance and support helped me in understanding the practical application of data analytics concepts.

I would also like to thank my peers and online resources that assisted me during the project. This project has helped me enhance my knowledge in data visualization, data analysis, and decision-making using tools like Power BI and Python.

2. Objective and Scope:

The main objective of this project is to analyse employee attrition using HR data and identify the root causes that lead to employees leaving the organization.

The scope of this project includes:

- Identifying key drivers of attrition
- Understanding employee behaviour patterns
- Building an interactive dashboard using Power BI
- Providing actionable recommendations to reduce attrition

3. Problem Statement:

Employee attrition is a serious issue for organizations as it leads to:

- Increased hiring and training costs
- Loss of experienced employees
- Decreased overall productivity

In this project, the problem is to:
Identify why employees are leaving and which factors are most responsible for attrition.

The focus is on analysing variables such as:

- Salary (Monthly Income)
- Overtime
- Job Satisfaction
- Work-Life Balance
- Years at Company

4. Existing Approaches

Organizations use traditional methods such as:

- Employee surveys
- Exit interviews
- HR reports

However, these methods lack clear insights and visualization.
Data analytics tools like Power BI provide better understanding through interactive dashboards.

5. Approach / Methodology:

The project was carried out using the following steps:

- Data collection from HR dataset
- Data cleaning using Python (Jupiter Notebook)
- Data preprocessing and transformation
- Data visualization using Power BI
- Analysis of patterns and trends

6. Workflow:

- Data Collection
- Data Cleaning
- Data Transformation
- Dashboard Creation
- Insight Generation
- Report Preparation

7. Assumptions:

- Dataset is accurate and reliable
- No major missing or incorrect values
- Employee data reflects real-world scenarios

8. Implementation:

The dataset was cleaned and processed using Python and then visualized in Power BI.
Dashboard Pages:

Page 1: Overview

- Total Employees
- Attrition Count
- Attrition Percentage
- Summary charts

Provides overall workforce understanding

Page 2: Root Cause Analysis

- Overtime vs Attrition
- Salary vs Attrition
- Job Satisfaction

Identifies main reasons for attrition

Page 3: Insights / Analysis

- Key patterns
- Employee behavior trends

Helps in decision making

9. Solution Design:

The solution is designed as an interactive dashboard with:

- KPI indicators
- Multiple pages (Overview, Root Cause, Insights)
- Filters for better analysis
- Clear and easy-to-understand charts

10. Challenges and Opportunities:

Challenges:

- Data cleaning complexity
- Identifying key relationships
- Designing effective visuals

Opportunities:

- Predictive analytics
- Advanced HR insights
- Real-time dashboard integration

11. Key Insights and Recommendations:

Insights:

- Employees working overtime show higher attrition
- Lower salary employees are more likely to leave
- Low job satisfaction leads to higher attrition
- Employees with fewer years at company leave more

Recommendations:

- Reduce overtime workload
- Improve salary structure
- Enhance employee engagement
- Improve work-life balance

12. Outcome / Conclusion:

A fully functional Power BI dashboard was developed to analyse employee attrition. The analysis revealed that overtime, salary, and job satisfaction are the main factors influencing attrition.
This dashboard helps organizations make better HR decisions and improve employee retention.

13. Enhancement Scope:

- Machine Learning for prediction
- Real-time data integration
- Advanced analytics

14. Research Questions and Responses

Question	Answer
Why do employees leave?	Due to overtime, low salary, dissatisfaction
Does salary impact attrition?	Yes, lower salary leads to higher attrition
What affects attrition most?	Overtime and job satisfaction

15. References

- HR Dataset
- Power BI Documentation
- Online learning resources